

Testing, data collection and analysis

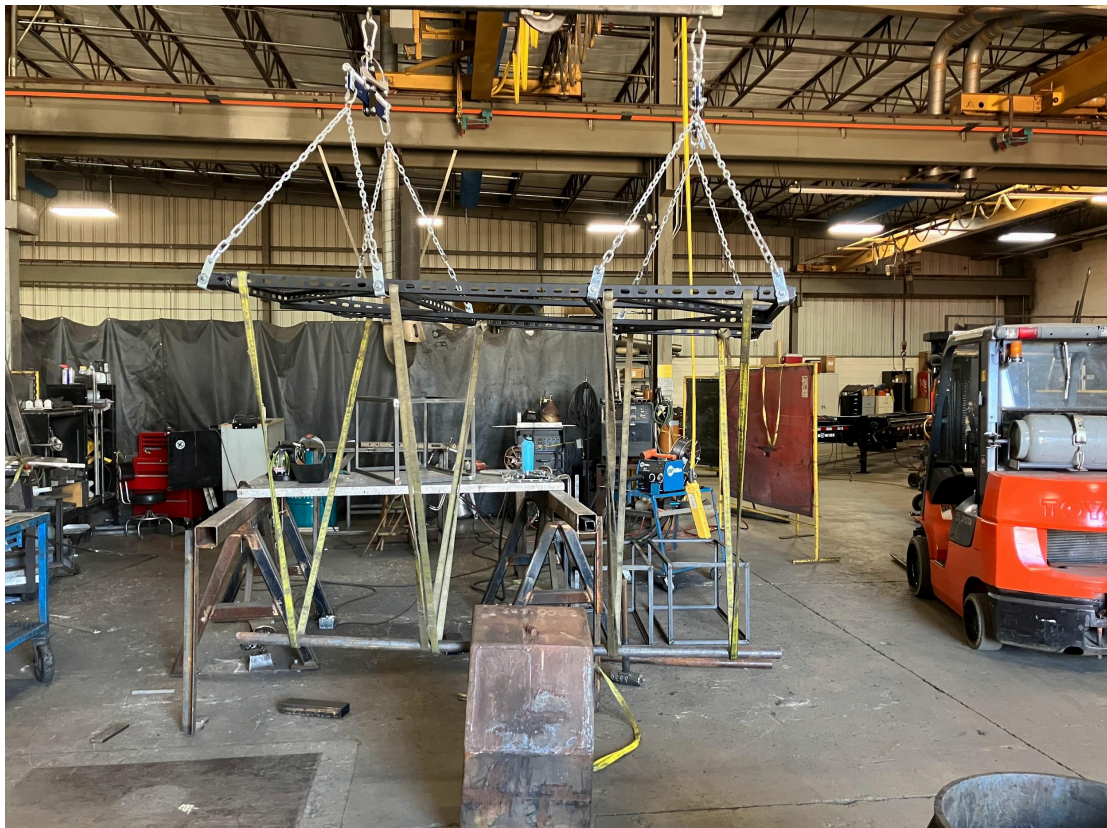
Through the conduct of several tests for high priority requirements that are reasonable based on instructional contexts, or through physical or mathematical modeling, the student demonstrates considerable understanding of testing procedure, including the gathering and analysis of resultant data; the analysis of the effectiveness with which the design met stated goals includes a consistently detailed explanation [and summary] of the data from each portion of the testing procedure and from expert reviews, generously supported by pictures, graphs, charts and other visuals; the analysis includes an overall summary of the implications of all data for proceeding with the design and solving the problem.

Procedure

Our testing was done in two phases: one to stress test the frame and one to check the leveling system, in that order.

Stress Testing

1. Hang the frame from 1 static point and the chain block with the engine levelers, ensuring that it is level on both axes:
2. Hang a 1-ton static weight off of the eight planned anchoring points as shown below:





3. let sit for 30 minutes
4. Measure maximum deflection and record. We utilized a straight aluminum bar to check for deflection (distance from reference to frame center) as pictured below on both axes.



This beam served as our point of reference to measure the distance the center of the frame deflected from the edges.

5. Release the static weight, and check for unloaded deflection off of horizontal (if unloaded deflection occurs, scrap the frame. Plastic deformation will interfere with the permanent integrity of the structure.)
6. Repeat steps 2-5 two more times

Leveling system testing (unweighted)

1. Hang the frame from 1 static point, and the chain block is retracted as far as possible
2. Run the chain block out as far as possible. Measure and record the time taken.
3. Bring the frame back to level, as it was during stress testing
4. Run the engine levelers till they bottom out on one side
5. Measure and record the angle the frame makes to the horizontal
6. Run the levelers all the way to the other side and measure the time taken

Data:

Long side deflection:

Trial 1: 0.46"

Trial 2: 0.51"

Trial 3: 0.49"

0.75 in (highest of all trials)

Short side deflection: not measurable (all trials)

Unloaded deflection: N/A (all trials)

Time to lower the chain block: 1:57

Maximum angle to Horizontal: 35 deg

Time for engine levelers: 28 s

Analysis:

The frame easily fulfills the strength requirements, with overhead for peace of mind. The leveling system is also functional, allowing 1 person to adjust the animal's tilt after it's in the air. In addition, all of this was done with unistrut, creating dozens of attachment points for tarps or other equipment.